

Table 5. Statistical Comparison of Primary Pharmacokinetic Parameters between Eksunbi and EU Stelara (Pharmacokinetic Analysis Set, Study SB17-1001)

PK Parameter	Treatment	N	n	Geometric LS Mean	Ratio A/B	90% CI of Ratio
AUC _{inf} (ng·h/mL)	SB17	67	62	4958100	0.99	0.90; 1.08
	EU Stelara	67	64	5020000		
C _{max} (ng/mL)	SB17	67	62	4882	0.90	0.82; 0.98
	EU Stelara	67	64	5433		

A = Eksunbi; B = EU Stelara

CI = confidence interval; Geometric LS Mean = geometric least squares mean; N = number of subjects in Pharmacokinetic Analysis Set; n = number of subjects in the analysis; PK Parameter = pharmacokinetic parameter Samples for PK analysis (a total of 23 PK samples from 23 subjects) were excluded due to major protocol deviation (centrifuged in low speed) from the PK parameters calculation.

The geometric LS Mean ratio (90% CI) of Eksunbi and EU sourced Stelara for AUC_{inf} and C_{max} were 0.99 (0.90 to 1.08) and 0.90 (0.82 to 0.98), respectively. These values for the 90% CIs were within the predefined equivalence margin of 0.8 to 1.25 however the geometric LS Mean ratio (90% CI) of Eksunbi and EU Stelara for C_{max} was 0.90 (0.82 to 0.98), which is within the pre-defined equivalence margin, but does not include 1.

The statistical comparison between Eksunbi and US Stelara and between EU Stelara and US Stelara showed that the corresponding 90% CIs of the geometric LS Mean ratios of the AUC_{inf} and C_{max} were within the pre-defined equivalence margin. Therefore, the PK profiles between Eksunbi and US Stelara and between EU Stelara and US Stelara were equivalent.

According to the Population-PK analyses performed in patients with PsO and PsA, patient's body weight is considered as the primary covariate affecting PK parameters of apparent clearance (CL/F), and apparent volume of distribution (V/F) of ustekinumab [Zhu et al., 2009; Zhu et al., 2010]. Generally, heavier patients tend to have lower serum ustekinumab concentrations at each dose [Lebwohl et al., 2010]. Based on the baseline comparability of demographics and baseline characteristics between the three treatment groups in Study SB17-1001, body weight was identified as a statistically different variable between the treatment groups (p-value = 0.0199).

Table 6. Baseline Comparability of Demographic and Baseline Characteristics between Main Treatment Group (Randomised Set, Study SB17-1001) (Ad-hoc Analysis)

Variables	Test	Degrees of Freedom (df)	Test Statistics	P-value
Age (years)	F-test	2	0.581	0.5604
Gender	Chi-squared test	2	0.042	0.9792
Height (cm)	F-test	2	0.290	0.7486
Body weight (kg)	F-test	2	3.996	0.0199
BMI (kg/m ²)	F-test	2	3.002	0.0520
Race	Fisher's exact test			0.8075
Ethnicity	Fisher's exact test			1.0000

BMI = body mass index

Accordingly, an ad-hoc analysis of covariance (ANCOVA) for the primary PK parameters with body weight as a covariate was performed after noting the 90% CI did not include 1.